Quantification: 682_z29
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	15
interior lacuna count	13
frame-edge lacuna count	2

Interior lacunae only: mean, median and sample SD (n = 13)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	1650.75	1511.5	948.377	px^2
major_axis_length_px	lacuna shape	74.0009	74.236	22.3194	px
minor_axis_length_px	lacuna shape	27.9394	29.9101	9.5919	px
aspect_ratio	lacuna shape	2.805	2.6218	0.775	unitless
eccentricity	lacuna shape	0.9144	0.9241	0.0615	unitless
solidity	lacuna shape	0.8593	0.8693	0.0551	unitless
roots_count	headline	7	7	3.4157	count
ring_length_r30_px	headline	299.615	299	121.517	px
ring_length_r60_px	reported	749.462	760	292.693	px
owned_length_px	ownership-dependent	2269.14	1782.67	2137.09	px
edge_count	ownership-dependent	99.5385	59	100.864	count
mean_edge_length_px	ownership-dependent	26.3152	24.6963	6.6832	px
perimeter_px	normalised	186.054	202.693	70.3766	px
ring_area_r30_px2	normalised	7571.77	7711	2170.77	px^2
ring_area_r60_px2	normalised	19073	19480	4883.71	px^2
in_frame_fraction_r30	normalised	0.9952	1	0.0173	unitless
in_frame_fraction_r60	normalised	0.9839	1	0.0537	unitless
ring_density_r30	normalised	0.0394497	0.0386337	0.0111315	px^-1
ring_density_r60	normalised	0.0390938	0.0376005	0.0096488	px^-1
roots_per_100px_perimeter	normalised	3.8347	2.9757	1.7283	per 100 px
ring_attached_length_r30_px	network v2	257.385	258	126.868	px
ring_attached_length_r60_px	network v2	560.769	471	341.595	px
ring_length_w_r30_px	network v2	335.973	343.806	132.849	px
ring_length_w_r60_px	network v2	842.489	848.169	326.431	px
sholl_crossings_r10	network v2	7.1538	8	2.7642	count
sholl_crossings_r20	network v2	9.8462	11	3.8264	count
sholl_crossings_r30	network v2	11	10	3.9158	count
ring_width_mean_r30_px	width	7.0396	7.1612	0.6108	px

Per field
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measure	value	units
analysed_area_px2	1.02273e+06	px^2
total_skeleton_length_px	36855	px
canalicular_length_density_per_px	0.036036	px^-1
junction_count	794	count
junction_density_per_px2	0.00077636	px^-2
lacuna_count	15	count
lacunae_per_px2	1.467e-05	px^-2
skeleton_component_count	246	count

mean_component_length_px	149.817	px
median_component_length_px	22	px
field_length_density_w_per_px	0.0407336	px^-1
field_density_without_flagged_per_px	0.0364744	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	513	px
unattached_component_count	32	count
canalicular_length_density_attached_per_px	0.036036	px^-1
field_length_density_with_unattached_per_px	0.0365376	px^-1

Canalicular width, the new measure

measure	value	units
width_mean_px	7.6752	px
width_median_px	8	px
width_p10_px	4	px
width_p90_px	10.198	px
width_pixels_used	34543	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	2083	86.508	34.896	2.479	0.915	0.8266	1.5435	13.4983	99.4023	True
2	2213	89.1614	32.1486	2.7734	0.9327	0.8778	-1.1291	134.914	921.292	False
3	446	49.6136	12.2658	4.0449	0.969	0.7866	-1.4445	249.987	449.359	False
4	3140	120.617	34.4194	3.5043	0.9584	0.8885	1.5609	431.048	220.093	False
5	2837	88.9183	41.496	2.1428	0.8844	0.9229	-1.5453	532.043	493.055	False
6	1634	75.9278	30.0551	2.5263	0.9183	0.8211	-1.4797	555.483	357.187	False
7	3781	121.501	43.6088	2.7862	0.9334	0.8771	1.4272	590.482	55.0987	True
8	1399	81.52	23.1918	3.515	0.9587	0.8541	1.5542	716.135	181.57	False
9	663	61.7115	15.8655	3.8897	0.9664	0.7408	-1.1603	728.596	861.06	False
10	513	41.8409	17.3844	2.4068	0.9096	0.8607	1.3405	830.565	75.5867	False
11	1130	64.3164	23.6688	2.7173	0.9298	0.8835	1.4062	842.292	125.237	False
12	1244	49.7173	33.3792	1.4895	0.7411	0.9325	-1.542	904.6	760.807	False
13	2966	92.1224	41.6334	2.2127	0.8921	0.895	-1.4899	920.401	307.385	False
14	1624	72.5441	29.7651	2.4372	0.9119	0.8476	-0.945	978.497	437.615	False
15								548.08	212.663	False

Network per lacuna

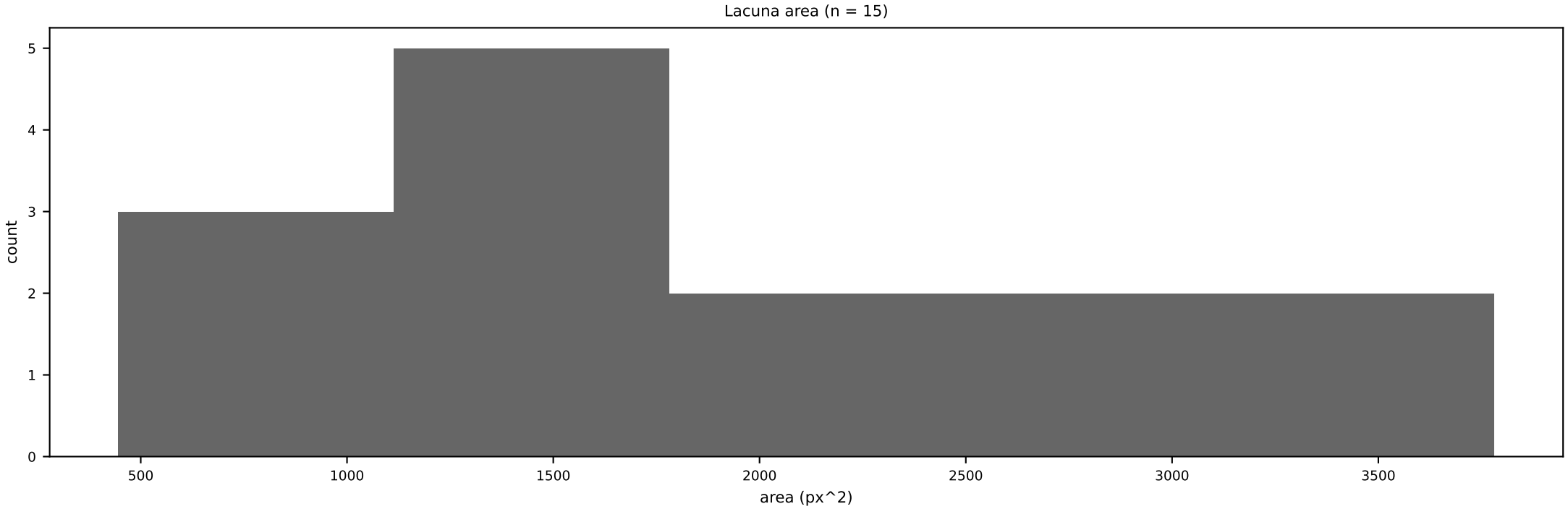
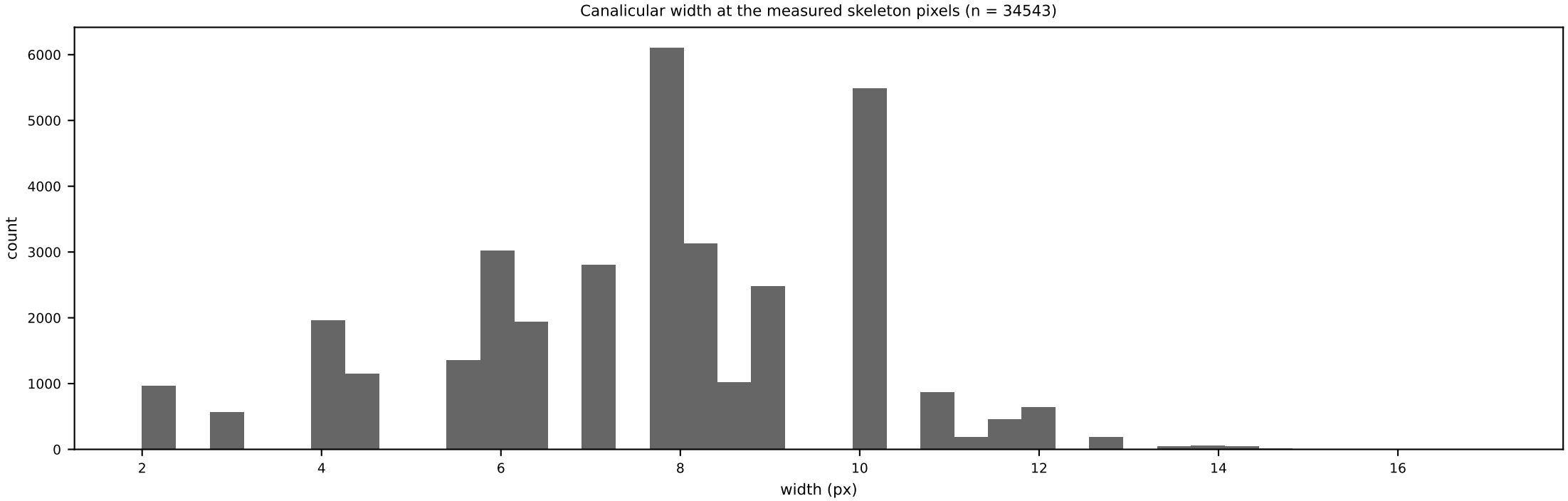
lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	3	128	230	90.8406	8	11.3551	
2	13	525	1298	6255.2	271	23.0819	7.3132
3	8	378	968	4890.6	238	20.5487	7.7186
4	8	429	1233	5608.21	275	20.3935	7.4282
5	7	292	823	3058.9	123	24.8691	7.0246
6	9	437	900	2676.87	118	22.6854	7.115
7	12	390	819	1415.5	63	22.4683	7.48
8	6	352	809	1782.67	59	30.2147	5.7984
9	8	208	584	2417.48	99	24.419	6.6552

10	6	164	455	444.534	18	24.6963	6.63
11	4	173	457	420.09	11	38.19	7.8561
12	2	158	443	404.333	16	25.2708	6.0685
13	7	317	760	889.625	36	24.7118	7.1723
14	12	299	473	608.257	29	20.9744	7.1612
15	1	163	540	42.0416	1	42.0416	7.574

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	262.096	5607	13819	0.5809	0.5584	0.0228286	0.0166437	1.1446
2	229.522	9151	23871	1	0.999	0.0573708	0.0543756	5.6639
3	120.811	6134	17911	1	1	0.0616237	0.054045	6.6219
4	309.279	11331	27269	1	1	0.0378607	0.0452162	2.5867
5	235.238	9387	21888	1	1	0.0311068	0.0376005	2.9757
6	214.167	8637	19737	1	1	0.0505963	0.0455996	4.2023
7	322.836	9940	21386	0.8402	0.7724	0.0392354	0.0382961	3.7171
8	229.48	8595	22042	1	1	0.040954	0.0367027	2.6146
9	156.853	6922	19429	1	1	0.0300491	0.0300582	5.1003
10	109.498	4245	11066	1	0.9865	0.0386337	0.0411169	5.4796
11	161.817	5491	13376	1	1	0.0315061	0.0341657	2.4719
12	147.196	6914	19480	1	1	0.0228522	0.0227413	1.3587
13	254.693	9728	23664	1	1	0.0325864	0.0321163	2.7484
14	202.693	7711	16371	0.9376	0.8055	0.0387758	0.0288925	5.9203
15	47.4558	4187	11845	1	1	0.03893	0.0455889	2.1072

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	20	20	145.397	253.652	1	4	4
2	488	1248	577.262	1467.61	11	17	18
3	357	968	428.049	1078.31	8	13	11
4	386	1048	480.848	1385.49	11	13	16
5	246	504	319.995	910.056	7	12	13
6	368	712	486.534	1007.52	9	11	15
7	377	767	438.978	923.311	10	12	12
8	304	471	387.693	893.269	9	12	10
9	188	407	244.794	679.583	5	7	8
10	149	330	183.681	506.99	6	4	5
11	173	404	199.095	511.505	5	7	8
12	112	272	179.924	510.63	5	6	8
13	258	487	353.421	848.169	8	12	14
14	290	403	343.806	538.73	8	9	7
15	27	36	182.539	614.487	1	5	10

682_z29: distributions, pre-validation, px